

20FF RS4 MACHINES PER TX SECONDARY**FEASIBILITY STUDY AND REPORT**

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1. INTRODUCTION

This purpose of this tech note is to describe the outcome of the GEV internal feasibility assessment of offering 2 RS4 machines per secondary winding of the GSUT, how GEV would propose to implement the control of this configuration of machines, the limitations of equipment ratings, and the impact on the short circuit contribution at the GEP which is the key technical evaluation criteria of the LT29 programme.

2. CONTROL OF MULTIPLE RS ON A SINGLE BUS

GEV has considered 2 control methodologies, which are both technically viable options that could be explored in detail on a project specific basis.

The main differences are summarised in Table 1 and described in further detail in sections 2.1 and 2.2 below.

Table 1- summary of control options

Option	Additional hardware required	Additional software required	Allows different AVR types
Basler cross current compensation	One additional CT per machine with shorting switch interlocked with GCB	None (but requires AVR set up)	No
Load balancing control in DCS	None	Additional simple control logic required	Yes

2.1 BASLER CROSS CURRENT COMPENSATION

Cross current compensation (reactive differential) mode in the Basler AVR controller allows for 2 or more paralleled generators to share a common load. As shown in Figure 1 each generator is controlled by a DECS-250 using the DECS-250 cross current compensation input (terminals CCCT+ and CCCT-) and a dedicated, external current transformer (CT) to sense generator current. The resistors shown in Figure 1 are used to set the burden and may be adjusted to suit the application. The power rating of the resistors would need to be sized for the specific application as appropriate.

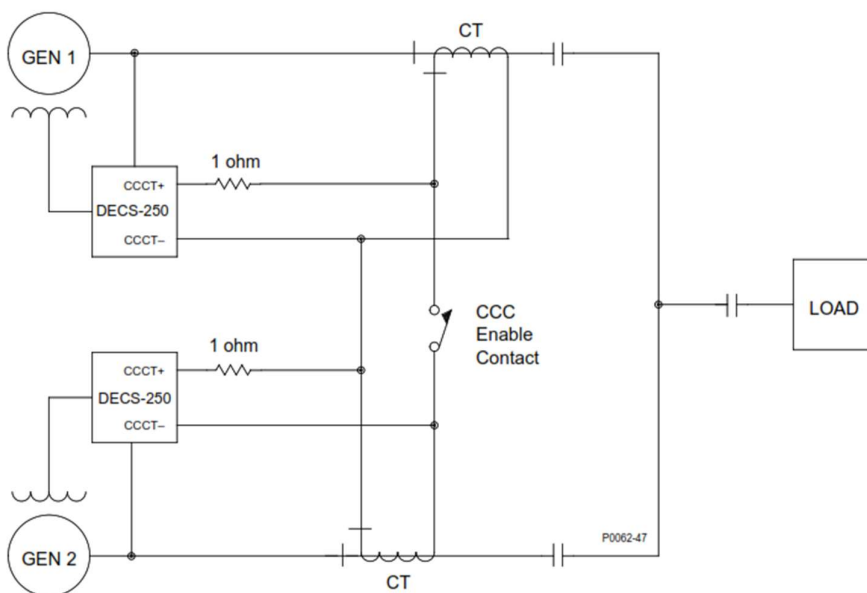


Figure 1 - connections for cross circuit compensation

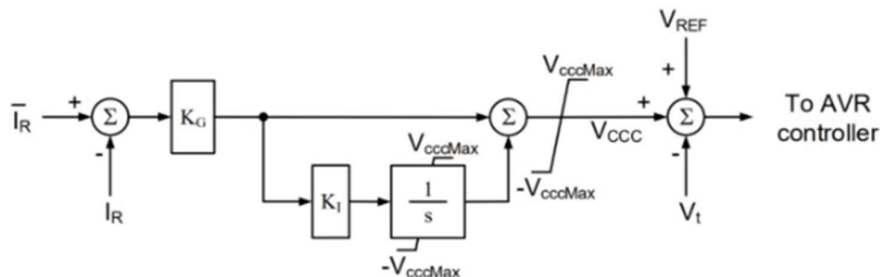


Figure 2 - proposed cross circuit compensation block diagram

The block diagram depicted in Figure 2 consists of a PI controller whose input is the difference between the reactive current of the target unit (I_R) and the average reactive current of all other paralleled units ($\bar{I}_R$). K_G and K_I are the proportional and integral gains of the PI controller respectively. If K_I is equal to zero, the block diagram will act like the old network load share scheme. The output of the block diagram is limited by the V_{ccMax} variable. The final output of the PI controller (V_{cc}) is passed to the AVR summing point of the DECS-250. Note, if a machine is offline, then its CT for cross current compensation must be shorted. Figure 3 shows a 0.2% VT error induced at 40s where 2 machines, RS01 and RS03, continue to share the load equally.

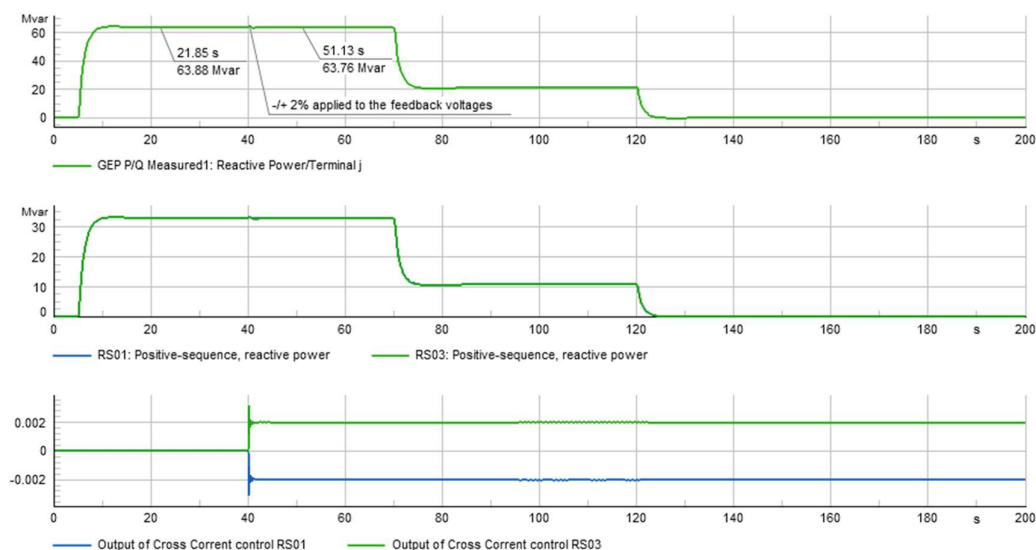


Figure 3 - 2 unit response to induced fault



2.2 LOAD BALANCING CONTROL IN DCS

Implementing the load balancing control in the DCS uses the same mechanism as the Basler cross current compensation, and can be implemented in the DCS without the need to include additional hardware (but additional software updates to the DCS are required). The Q feedback of each RS4 machine is compared to the average Q, and the difference creates a local voltage trim signal for load balancing. One of the advantages of this method is that it would be a common implementation regardless of the model of AVR controller used.

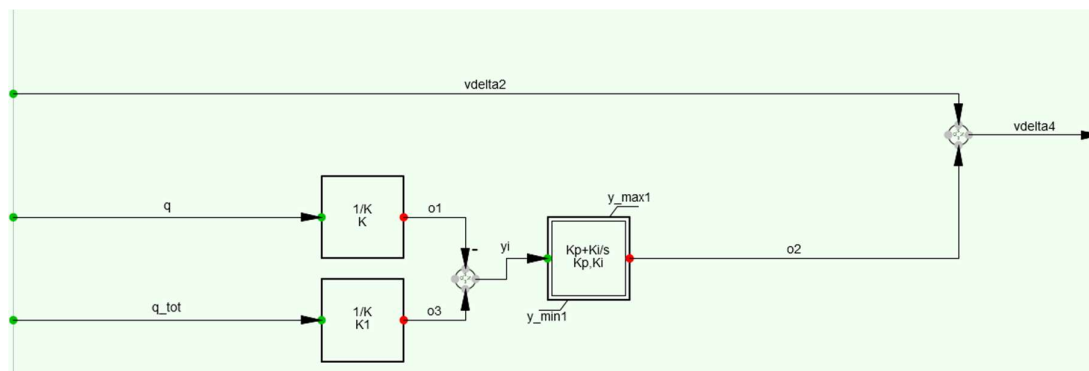


Figure 4 - proposed load balancing in DCS

Figure 5 shows a 0.5% VT error induced with load balancing control implemented in the DCS where 2 machines attached to the same transformer secondary winding, RS01 and RS03, continue to share the load equally.

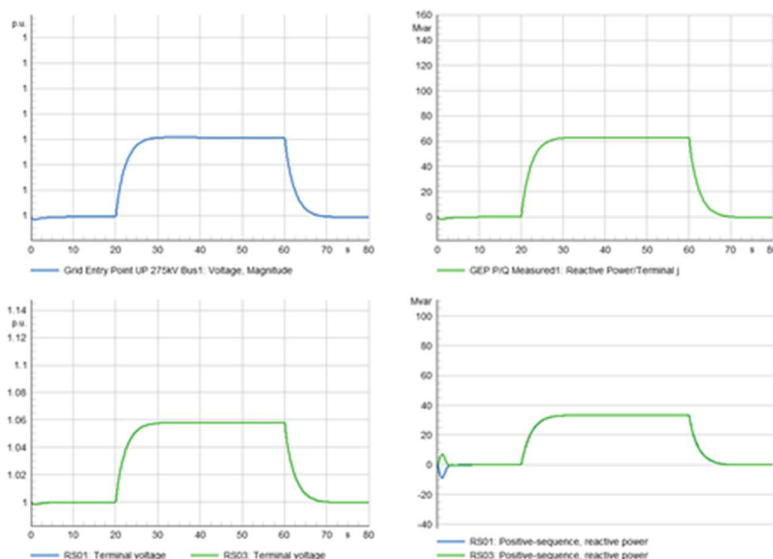


Figure 5 - VT error with load balancing control

Figure 6 shows the same 0.5% VT error induced without load balancing control implemented in the DCS where 2 machines attached to the same transformer secondary winding, RS01 and RS03, fight against each other.

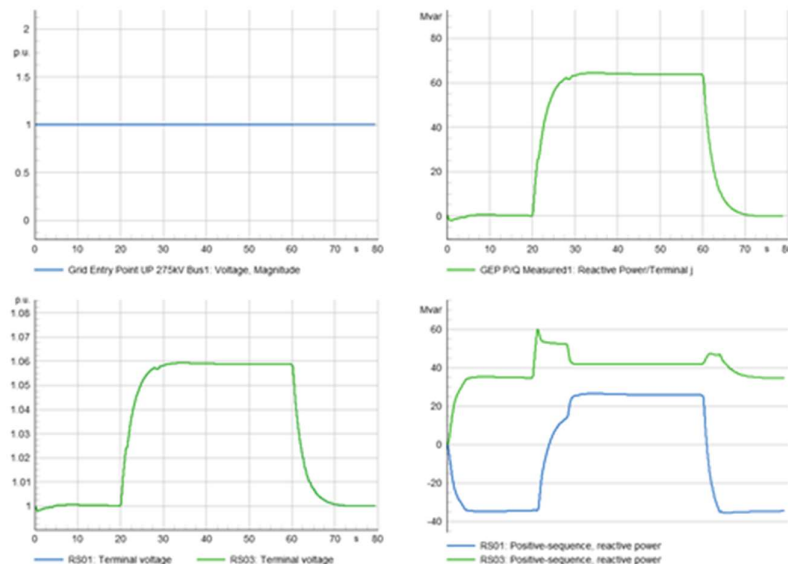


Figure 6 - VT error without load balancing control

3. EQUIPMENT RATING MAX SHORT CIRCUIT

GEV has (in the first instance) limited the transformer rating of this offering to 200/100/100 MVA with a HV-LV impedance of 8%. This would give a preliminary maximum peak current of 199kA (through current) and maximum breaking current 65kA (through current) through the GCB, which is within the current GCB ratings (220kA making and 80kA breaking). Bus-RS01 (as shown in Figure 7) would have I_{kss} 98kA, I_p 261kA with 2off RS4 machines per transformer secondary, an increase from I_{kss} of 63kA, I_p 166kA with 1off RS4 machine per secondary. The LV switchboard rating would increase to I_{kss} 31.6kA, I_p 68kA with 2off RS4 machines per transformer secondary, from I_{kss} of 31.5kA, I_p 70.2kA with 1off RS4 machine per secondary.

GEV's analysis is based on our PowerFactory DigSILENT models of the system, an overview of which can be found in Figure 7.

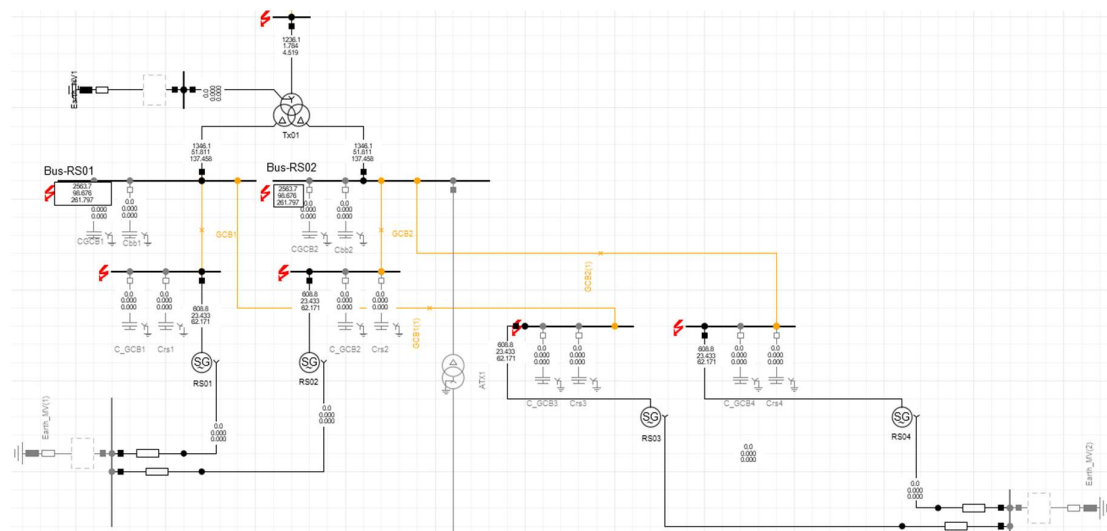


Figure 7 - system fault current analysis

4. COMPARISON TO 4OFF RS4 MACHINES ON 2OFF TX

Table 2 describes a preliminary overview of the short circuit contribution provided by each individual RS4 machine, with a single machine per 60MVA Tx secondary (Option 1 - GEV's standard offer for LT29), a single machine per 80MVA Tx secondary (Option 2 - GEV's offer for increased short circuit contribution per machine) and each of 2 machines on a 100MVA secondary having an assumed 50MVA of the secondary winding per machine, which is GEV's offering described in this document (Option 3).

Table 2 - Comparison of SCC

Machine Data			Option 1	Option 2	Option 3	Notes
Rating over excited	P_1	MVA _r	42.5	42.5	42.5	
Volts	V_1	V	15000	15000	15000	
Sync. Reactance	X_d	pu	1.103	1.103	1.103	
Transient reactance (sat)	X'_d	pu	0.115	0.115	0.115	
Sub-transient reactance (sat)	X''_d	pu	0.07	0.07	0.07	
SC transient time const (sat)	T'_d	s	0.814	0.814	0.814	
SC sub-transient time const (sat)	T''_d	s	0.047	0.047	0.047	
Time after fault	t	s	0.04	0.04	0.04	
Transient exponent	$\exp(-t/T'_d)$		0.9520	0.9520	0.9520	
Sub-transient exponent	$\exp(-t/T''_d)$		0.4270	0.4270	0.4270	
Constant SCL	P	MVA	39	39	39	$=P_1/x_d$
Transient SCL	P'	MVA	370	370	370	$=P_1/x'_d$
Instantaneous SCL	P''	MVA	607	607	607	$=P_1/x''_d$
SCL at time t at machine terminals	P_t	MVA	455	455	455	$= P + (P' - P)\exp(-t/T'_d) + (P'' - P')\exp(-t/T''_d)$
SCL at time t at machine terminals (IEC tolerances)			396	396	396	
TX rating						
Transformer Rating (secondary MVA per machine)	T	MVA	80	60	50	
Transformer Impedance	T_x	pu	0.08	0.08	0.08	
Transformer SC	T_{SC}	MVA	1000	750	625	$=T_x / T_{SC}$
SCL at 0 secs at PCC	SCL_{v2}	MVA	378	336	308	$=1/(1/P''+1/T_{SC})$
SCL at time t at PCC (HV₂) (data sheet values)	SCL_{v2} at t	MVA	313	283	263	$=1/(1/P_t+1/T_{SC})$
SCL at time t at PCC (HV₂) (+ IEC Tolerances)	SCL_{v2} at t	MVA	276	251	233	$=1/(1/P_{2t}+1/(T_{SC}/1.1))$

5. MODIFICATION RECORD

VERSION	DATE	AUTHOR	DETAILS
000	19 th January 2026	Matthew Toms	Initial release.